

Shubh Somani

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Summary — Self-taught artificial intelligence (AI) engineer and third-year computer science student with deep expertise building production-grade retrieval-augmented generation (RAG) systems, computer vision (CV) pipelines, and full-stack AI applications. Engineered 28+ GitHub repositories spanning multi-service large language model (LLM) orchestration, LangGraph state machines, and self-hosted infrastructure serving 10+ live services across 5 custom domains with zero recurring cost. Combines strong Python/TypeScript engineering with hands-on DevOps (Docker, Proxmox, Coolify) to ship real products from bare metal to browser.

Education

Manipal University Jaipur

B.Tech Computer Science (Data Science)

GD Goenka Public School

CBSE 12th (PCM)

Certifications

- Microsoft Azure - From Zero to Hero
- Stanford - Unsupervised Learning, Recommenders, Reinforcement Learning
- Stanford - Supervised Machine Learning: Regression and Classification

Skills

Languages	Python, TypeScript, JavaScript, Java, C, Bash	Frameworks	FastAPI, Django, React, Next.js, Streamlit, Typer
ML/AI	PyTorch, LangChain, LangGraph, Sentence-Transformers, Scikit-Learn, XGBoost	Infrastructure	Docker, PostgreSQL, Nginx, Proxmox, Coolify, Tailscale, Cloudflare
RAG & Vector	pgvector, ChromaDB, LlamaIndex, BGE-M3, BM25, Cross-Encoders, HyDE	LLM Providers	Groq, OpenRouter, Google Gemini, Ollama, NVIDIA NIM
Computer Vision	YOLOv8, MediaPipe, OpenCV, EasyOCR, SORT Tracking		

Experience

IEEE Computer Society

Oct 2024 – Apr 2026

Technical Core Team Member

- Architected a full-stack retrieval-augmented generation (RAG) hiring platform — 17 modules across 8,500 lines — featuring hybrid retrieval fusion combining dense vector search, BM25 keyword search, and HyDE hypothetical document queries via Reciprocal Rank Fusion (RRF) with cross-encoder re-ranking.
- Implemented a defense-in-depth safety system with 3 independent validation layers: 19-pattern regex guardrails (prompt injection, jailbreak, command injection), LLM gatekeeper (pre-generation context sufficiency), and LLM auditor (post-generation faithfulness), validated through 12 adversarial red-team tests with over 80% attack block rate.
- Engineered 20+ document format ingestion pipeline (PDF, DOCX, PPTX, XLSX, EPUB, HTML, images) through marker-pdf and Docling auto-routing with heading-aware chunking and LLM-generated chunk summaries.

Projects

YouTube-to-Posts — AI Content Pipeline

May 2026

- Architected a multi-service SaaS pipeline (5,900 lines) in 7 days for a 100k-subscriber YouTube creator that ingests YouTube channels, transcribes bilingual Hindi-English (Hinglish) content via Groq Whisper large-v3, profiles creator voice across 9 style dimensions through 8 parallel LLM passes, and generates platform-specific content (LinkedIn posts, Twitter threads) in the creator's exact language mix.
- Implemented two LangGraph generation modes — critique-refine (single draft with LLM critique) and n-variants-judge (4 parallel variants with LLM judge and quality gate) — with PostgresSaver checkpointing for resume on failure.
- Designed SaaS migration seams from day one: file-system behind OutputWriter protocol, all LLM calls through single LLMRouter chokepoint, Langfuse observability, and env-driven config — enabling multi-tenant migration without architectural rewrites.
- Built captions-first ingest strategy achieving 70-90% zero-cost video processing via public YouTube caption API, with yt-dlp audio fallback in isolated multiprocessing spawn workers to prevent memory leaks.

FullRag-Plus — Production RAG Platform

Apr 2026

- Engineered an enterprise-grade RAG platform (16,200 lines, 85 commits, 14 days) with FastAPI backend, React + TypeScript frontend, and PostgreSQL + pgvector (HNSW, 768-dim cosine) — serving 4 document loaders (PDF/DOCX/HTML/MD), structure-aware chunking with section-path tracking, and LLM enrichment (summaries, keywords, hypothetical questions).

- Implemented continuous evaluation with regression detection — APScheduler background job running daily retrieval metrics (Precision@k, Recall@k, MRR, NDCG@k) and LLM-as-judge scoring across 4 dimensions (faithfulness, relevance, completeness, coherence) with $\leq 10\%$ precision drop alerts.
- Built a dual-provider generation pipeline (Google Gemini, Groq) with auto-prompt selection (QA/INSUFFICIENT/TRANSCRIPT_FALLBACK based on retrieval score), TTL response cache (256-entry LRU), and transcript fallback for low-confidence queries.
- Containerized 3-service Docker Compose stack (pgvector, FastAPI, nginx + React SPA) with Alembic migrations and multi-stage Dockerfiles; shipped Streamlit admin dashboard with 13 debug pages for every pipeline stage.

HappyNest Website — Luxury Farm Stay

Apr – May 2026

- Developed and deployed a full-stack luxury farmstay website (happynestfarm.in) using Next.js + TypeScript with a canvas-scrubbed 91-frame GSAP ScrollTrigger video hero, 12 content sections, dark/light mode, and WhatsApp booking funnel — serving real guests at zero hosting cost on self-hosted homelab.
- Engineered an AI chat concierge with FastAPI server-side events (SSE) streaming, dual-LLM fallback (OpenRouter Gemini Flash Lite primary → NVIDIA NIM MiniMax M2.7), and Supabase persistence for chat sessions and lead capture — deployed via Coolify CI/CD through FRP tunnel to Oracle Cloud VPS.
- Implemented cache-augmented generation (CAG) approach with full knowledge base (~15K chars) injected into system prompt, eliminating vector database latency while covering 52 FAQ entries and complete property catalog.

Self-Hosted Infrastructure

2024 – Present

- Architected and maintain a complete homelab ecosystem: Proxmox hypervisor (6-year-old pandemic PC) running Docker LXC with Coolify PaaS, FRP tunnels through Oracle Cloud free-tier VPS for public ingress, Tailscale mesh VPN for inter-device communication, and Cloudflare Tunnel for media-heavy services — supporting 10+ services across 5 domains with zero monthly hosting cost.
- Deployed pgvector databases, self-hosted Supabase, Ollama local LLM inference, multiple embedding models (BGE-M3, sentence-transformers, nomic-embed-text), and full observability stack (Langfuse, LangSmith) — enabling production AI workloads entirely on repurposed consumer hardware.

Interactive 3D Portfolio

Apr 2026

- Built an interactive 3D portfolio at shubhsomani.tech using React + TypeScript + Vite with Spline 3D character, Rapier physics gravity-ball simulation (30 textured spheres with mouse-driven kinematics), GSAP ScrollTrigger animations, custom cursor with spring physics, and 1,900 lines of hand-crafted CSS.

Computer Vision Toolkit

Aug 2024 – Jan 2025

- Developed a multi-module CV toolkit covering pose estimation (MediaPipe BlazePose, 33 landmarks), license plate recognition (custom YOLOv8 + SORT tracker + EasyOCR with Indian-format validation), and object detection/counting (YOLOv8l with line-cross detection and region masking).